Jiayi Liwei

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Education

University of California, Los Angeles September 2024–Present

B.S. in Applied Mathematics Cumulative GPA: 3.64/4.00

Selected coursework: Linear algebra, real analysis, complex analysis, differential equations, and dynamical systems.

In progress: Abstract algebra; ordinary differential equations; Foundations of Machine Learning and AI (graduate level).

Research Experience

Finite-Time Dynamics in FPUT and Toda Lattices January 2026–Present

Independent research following UCLA MATH 134

- Investigate energy redistribution in Fermi-Pasta-Ulam-Tsingou oscillator chains using fourth-order Yoshida symplectic integration in Python.
- Compare spectral entropy, finite-time Lyapunov estimates, and drift of a selected Toda integral using diagnostic-specific references and numerical controls.
- Analyze lattice-size dependence and sensitivity to timestep and tangent-vector initialization; assess differences between diagnostics within finite observation windows.
- Conduct FPUT–Toda switching experiments with paired FPUT controls to examine the effects of intermediate Toda evolution on subsequent FPUT dynamics.

Research code: github.com/Argonriel/FPUT_Symplectic_Integrator

3D Reconstruction

Collaborative research; second author on a manuscript in preparation

- Reviewed and synthesized related literature on 3D reconstruction.
- Drafted the manuscript's introduction, literature review, preliminaries, and conclusion.

Teaching Experience

UCLA Olga Radko Endowed Math Circle 2025–2026 academic year

Instructor

Taught weekly problem-solving classes for the AMC 10/12 and AIME.

Honors and Awards

Dean's Honors List, UCLA Winter 2026

Valedictorian, Wycombe Abbey School Hangzhou 2024

Academic Activities

Women in Math Mentorship Program, UCLA Fall 2026–Present

Mentee

Technical Skills

Scientific computing: Python (NumPy, SciPy, Numba), Matplotlib, symplectic integration, numerical sensitivity analysis.

Research tools: Git, LaTeX, Linux/Unix.